Off-centre CoE cosmology: the mathematical workflow from geometry to the Planck TT spectrum

Working notes for C. A. (revision 3, compiled 23 October 2026)

Abstract

These notes take each step of the off-centre centre-of-expansion (CoE) workflow and derive it: inputs, intermediate steps, the meaning and units of each term, a numerical example using the values actually used in the code, and the conditions under which each approximation holds or fails. The order is: geometry and the radiation/matter history; density laws and the CatWISE lock used in the Migkas fit; spherical harmonics, C_ℓ, D_ℓ and cosmic variance; a realistic void/cluster web and the Monte-Carlo kernel ladder; the stochastic density field, Poisson's equation, the moving-lens source and its exact and Limber projections; the fixed low-ℓ background $D_\ell^{\rm fix}$; the acoustic and damped half-scale fits; and the residual decomposition. Goodness of fit is reported only as χ^2 and Pearson r^2, both defined in §4. These notes contain no pass/fail verdicts. Where a figure carries such a label, it is described as a label produced by the plotting script. Items that could not be traced to code or data are marked [unverified: …].

1 Geometry and time dependence

Set-up and symbols. D: distance from the observer to the centre of expansion (CoE); working value $D = 58.2\,\mathrm{Gpc}$. R: the observer's vision radius, i.e. the comoving distance to the CMB source surface, explored at $R = 4$, 8, 14 Gpc. Outer edge of the matter: $1.25D$ (the figure uses 72.83 Gpc). No cosmological constant; dynamics are matter plus early radiation.

Volumes (worked). A sphere of radius r has $V = \frac{4}{3}\pi r^3$. Observer sphere, $R = 4$: $V_R = 4.18879 \times 64 = 268.1\,\mathrm{Gpc}^3$. Sphere to $D + R = 62.2$ Gpc: $4.18879 \times 62.2^3 = 1.008 \times 10^6\,\mathrm{Gpc}^3$, so $V_R/V_{D+R} = 268.1/1.008 \times 10^6 = 0.0266\%$. Outer sphere 72.83 Gpc: $1.618\times10^6\,\mathrm{Gpc}^3$, so $V_R/V_{\rm out} = 0.0166\%$. (Arithmetic checked here; the 72.83 vs $1.25 \times 58.2 = 72.75$ difference is in the source figure.)

Radiation and matter densities. Inputs: $a(t)$, the scale factor ($a = 1$ today), and $a_{\rm eq} = 1/3400$. *Step 1*: matter is conserved particles in a volume $\propto a^3$, so $\rho_m = \rho_{m,0}a^{-3}$. *Step 2*: photons are also diluted as a^{-3}, but each photon's energy $h\nu \propto a^{-1}$ (redshift), so $\rho_r \propto a^{-4}$. *Step 3*: write $\rho_r = \rho_{r,0}a^{-4}$ and define equality by $\rho_r(a_{\rm eq}) = \rho_m(a_{\rm eq})$. Then $\rho_{r,0} = a_{\rm eq}\rho_{m,0}$, so

$$\frac{\rho_r}{\rho_{m,0}} = \frac{a_{\rm eq}}{a^4}, \qquad \frac{\rho_m}{\rho_{m,0}} = \frac{1}{a^3}. \tag{1}$$

Example: at $a = 10^{-4}$, $\rho_r/\rho_{m,0} = 2.94 \times 10^{12}$ and $\rho_m/\rho_{m,0} = 10^{12}$, so radiation dominates by a factor of $a_{\rm eq}/a = 2.94$.

From densities to $a(t)$. The Friedmann equation without Λ or curvature is $H^2 \equiv (\dot a/a)^2 = \frac{8\pi G}{3}(\rho_r + \rho_m)$. Substituting Eq. (1) gives $H = H_*\sqrt{a_{\rm eq}a^{-4} + a^{-3}} = H_* a^{-2}\sqrt{a_{\rm eq} + a}$ with $H_*^2 = 8\pi G\rho_{m,0}/3$. Since $dt = da/(aH)$,

$$H_* t(a) = \int_0^a \frac{a'\,da'}{\sqrt{a_{\rm eq} + a'}} = \tfrac{2}{3}(a - 2a_{\rm eq})\sqrt{a + a_{\rm eq}} + \tfrac{4}{3}a_{\rm eq}^{3/2}. \tag{2}$$

The intermediate step is the substitution $u = a' + a_{\rm eq}$, giving $\int (u - a_{\rm eq})u^{-1/2}du$. H_* is chosen so that $t(1) = 13.8\,\mathrm{Gyr}$; this is an assumption, not an output. Plugging in: $t(a_{\rm eq}) = 40.8\,\mathrm{kyr}$ and $t(1/1090) = 297.5\,\mathrm{kyr}$ (recombination). Limits: for $a \ll a_{\rm eq}$, $H_* t \approx a^2/(2\sqrt{a_{\rm eq}})$, so $a \propto t^{1/2}$; for $a \gg a_{\rm eq}$, $H_* t \approx \frac{2}{3}a^{3/2}$, so $a \propto t^{2/3}$.
Validity. Standard homogeneous FRW dilution. It is put in by assumption for the CoE universe. It would fail if the CoE dynamics are not FRW-like (inhomogeneous LTB shells evolve at different rates) or if $D(t)$ from the Newtonian integrator differs; that replacement is listed as future work.

Volume growth rates. If a length $\propto t^p$, then $V \propto t^{3p} \equiv t^n$ and $\dot V/V = n/t$. Radiation era $p = \frac{1}{2}$, so $n = \frac{3}{2}$. For D in the matter era $p = \frac{2}{3}$, so $n = 2$; today $2/13.8\,\mathrm{Gyr} = 0.145\,\mathrm{Gyr}^{-1}$. For the horizon $R \propto t$, so $n = 3$.

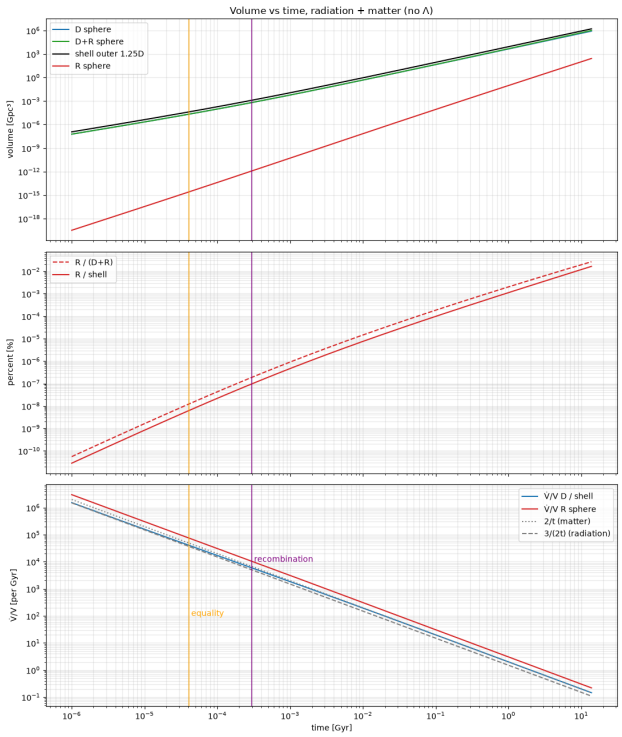


Figure 1: Volume versus time with radiation + matter (no Λ). Top: volumes of the D sphere, $D{+}R$ sphere and outer sphere from $a(t)$ of Eq. (2). Middle: V_R as a percentage of the larger volumes. Bottom: $\dot{V}/V = n/t$. Vertical lines: equality (41 kyr), recombination (297 kyr).

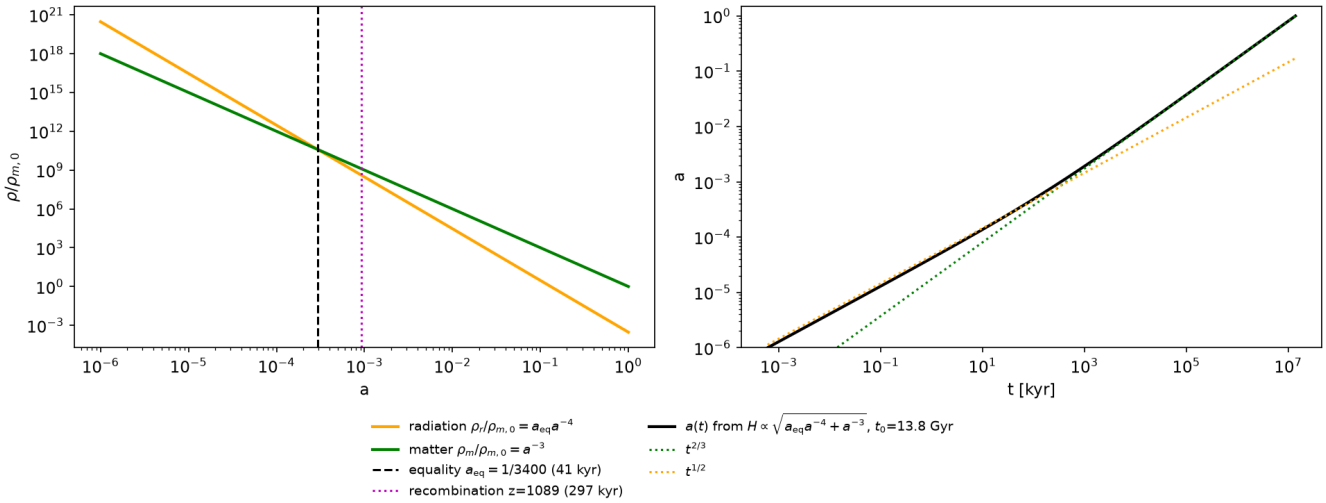


Figure 2: Eq. (1) (left, units of today's matter density) and Eq. (2) (right) for $a_{\rm eq} = 1/3400$, $t_0 = 13.8\,$Gyr. Dotted: the limits $t^{1/2}$ and $t^{2/3}$.

2 Density laws, the CatWISE lock and the Migkas fit

Observable. Migkas et al. measure the galaxy-cluster scaling-relation normalisation A in sky regions relative to its all-sky value, $y_i = A/A_{\rm all}$, in 71 bins of galactic longitude l with errors σ_i. In the CoE picture, a line of sight at angle ψ to the CoE axis crosses a different matter density and recession velocity from one at $\pi - \psi$, which produces a dipole-like modulation.

Line-of-sight geometry. Put the CoE at the origin and the observer at $\mathbf{D}$. A point at distance s from the observer in direction $\hat{n}$ has $r(s,\hat{n}) = |\mathbf{D} + s\hat{n}| = \sqrt{D^2 + s^2 + 2Ds\cos\psi}$, where $\cos\psi = \hat{D}\cdot\hat{n}$. Averaging

a response f over the cluster depth distribution $p(s)$,

$$\frac{A}{A_{\rm all}}(\hat{n}) = \frac{\int ds\, p(s)\, f[\rho(r(s,\hat{n}))]}{\left\langle \int ds\, p(s)\, f[\rho] \right\rangle_{\rm sky}}. \tag{3}$$

The denominator normalises the all-sky mean to 1. Because the scale lengths below are $L \gg s$, expanding to first order gives $\rho(r) \approx \rho(D)[1 + (\rho'/\rho)_D\, s\cos\psi]$: a pure dipole in $\cos\psi$ with amplitude set by the local logarithmic gradient. [unverified: the exact form of f and $p(s)$ in the Migkas scripts was not re-derived here].

CatWISE lock (worked derivation). Inputs: shape $g(x)$, $x = r/L$; $D = 58.2\,$Gpc; vision radius $d = 14.3005\,$Gpc; $\Omega_m = 0.3$, $H_0 = 70$. *Step 1, normalisation*: $\rho(r) = \rho_0 g(r/L)$ with $\rho(D) = 0.3\rho_{\rm crit}$, so $\rho_0 = 0.3\rho_{\rm crit}/g(D/L)$. *Step 2, contrast*: require

$$\delta \equiv \frac{\rho(D-d) - \rho(D+d)}{\rho(D)} = 0.031. \tag{4}$$

This is twice the CatWISE number-count dipole $\mathcal{D} = 0.01554$ (Secrest et al. 2021; $2\mathcal{D} = 0.03108$, rounded). The factor 2 converts a dipole amplitude (peak-to-mean) into the peak-to-peak difference. *Step 3, exponential*: $g = e^{-x}$ gives $\rho(D \mp d)/\rho(D) = e^{\pm d/L}$, so $\delta = e^{d/L} - e^{-d/L} = 2\sinh(d/L)$ and

$$L = \frac{d}{\operatorname{arcsinh}(\delta/2)} = \frac{14.3005}{0.0154994} = 922.65\ \text{Gpc}. \tag{5}$$

Then $L/D = 15.85$, $\rho_0/\rho_{\rm crit} = 0.3\,e^{58.2/922.65} = 0.3195$, and the local gradient is $\rho'/\rho|_D = -1/L = -0.001084\,\text{Gpc}^{-1}$. With $d = 3\,$Gpc instead, $L = 193.56\,$Gpc. *Step 4, other shapes*: Eq. (4) is solved numerically for L with the shape parameters held fixed (table). Every law reproduces $\delta = 0.031$ to 10^{-15}.

law	$g(x)$	L [Gpc]	$\rho_0/\rho_{\rm c}$
exp	e^{-x}	922.6	0.3195
power	$(1+x)^{-n}$, n=1.20	1049.3	0.3201
cored	$(1+x^2)^{-1}$	321.9	0.3098
Gauss	$e^{-x^2/2}$	231.5	0.3096
soft	$(1+x^\alpha)^{-\beta}$, α=0.30, β=1	5109.6	0.3784
stretched	e^{-x^η}, η=0.25	15246	0.3847
η-edge	e^{-x^η}, η=0.0500	0.849	1.0319

Meaning: across $D \pm d$ all laws have $\rho'/\rho \approx -0.00106$ to $-0.00109\,\text{Gpc}^{-1}$. The lock fixes the local gradient, which is what the dipole measures, so inside the window the laws are indistinguishable (Fig. 3). They differ only near the CoE, where no data constrain them. The level-set scan imposed $\eta \geq 0.25$; the η-edge solution comes from a separate free-η fit. Sources: `catwise-density-levelset-fit.json`, `catwise-eta-migkas-fit.json`.

Validity. The first-order expansion needs $d \ll L$. For exp, $d/L = 0.0155$, so it holds. For the η-edge, $L = 0.849\,\text{Gpc} \ll d$, so the law is far from linear across the window, even though δ is still met exactly.

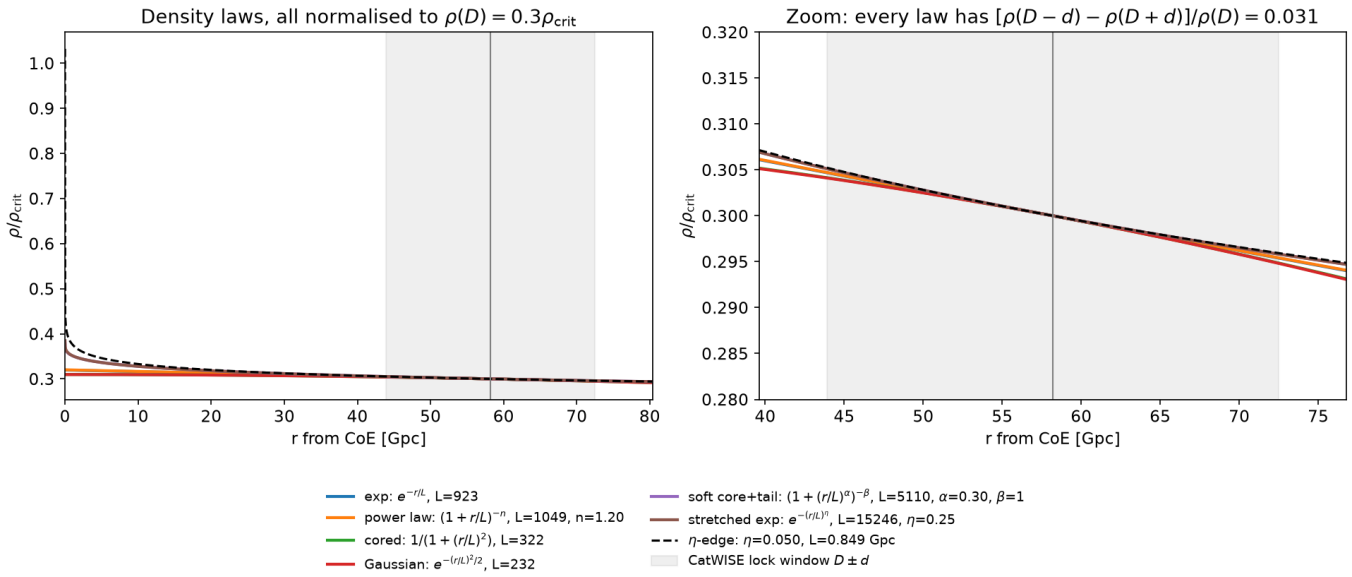


Figure 3: CatWISE-locked laws $\rho/\rho_{\rm crit}$ versus distance from the CoE (left) and zoom on $D \pm d$ (right), plotted from `catwise-density-levelset.csv` and `catwise-eta-density.csv`. $D = 58.2\,$Gpc (line), $d = 14.3005\,$Gpc (band).

Fit. $\chi^2 = \sum_{i=1}^{71}(y_i - m_i)^2/\sigma_i^2$ (§4). The null model $m_i = 1$ gives $\chi^2 = 45.14$. At fixed axis longitude $l_0 = 15.2°$, with D refitted (so not 58.2) and L locked: exp 11.61 ($D = 48.26$), power 11.58 ($D = 47.86$), soft 10.43 ($D = 39.11$), stretched η=0.25 10.39 ($D = 38.47$). With l_0 free ($\approx 21.7°$): cored 11.60 ($D = 59.5$), Gauss 11.61 ($D = 59.8$). Free η-edge: 9.28 ($D = 34.96$, $l_0 = 19.9°$). The direct fits of Fig. 4 give 10.79 ($l_0 = 100°$) and 7.67 ($l_0 = 15.6°$), the latter equal to a reference sinusoid.

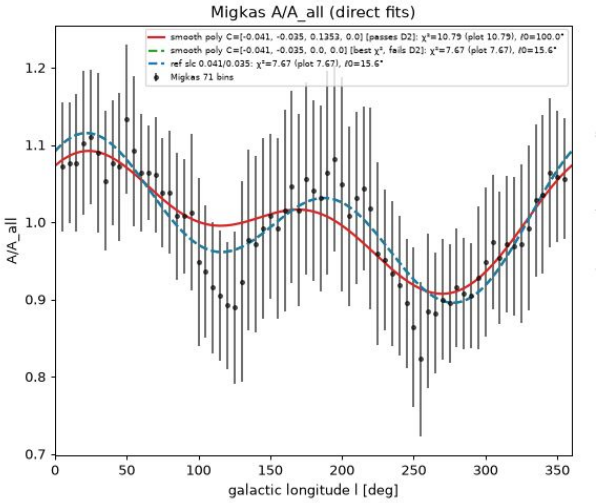


Figure 4: Migkas $A/A_{\rm all}$ (71 bins, black) with CoE direct fits. Red: smooth polynomial in $\cos\psi$, $\chi^2 = 10.79$, $l_0 = 100°$. Green dashed: best variant, $\chi^2 = 7.67$, $l_0 = 15.6°$. Blue dashed: reference sinusoid (amplitudes 0.041/0.035), $\chi^2 = 7.67$. The two humps come from $\cos\psi(l)$ at fixed galactic latitude.

3 Spherical harmonics, C_ℓ, D_ℓ and cosmic variance

Expansion. Input: a sky map $\Theta(\hat{n}) \equiv \Delta T/T$ (dimensionless). Since the $Y_{\ell m}$ are orthonormal on the sphere,

$$\Theta(\hat{n}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{n}), \quad a_{\ell m} = \int d\Omega\, \Theta Y^*_{\ell m}. \qquad (6)$$

ℓ is the number of nodal lines; a mode of degree ℓ has angular half-wavelength $\theta \approx 180°/\ell$, so $\ell = 220$ gives $0.82°$. For a statistically isotropic sky, $\langle a_{\ell m} a^*_{\ell' m'} \rangle = C_\ell \delta_{\ell\ell'} \delta_{mm'}$, and the estimator from one sky is $\hat{C}_\ell = \frac{1}{2\ell+1} \sum_m |a_{\ell m}|^2$.

From C_ℓ to D_ℓ. The map variance is $\langle \Theta^2 \rangle = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell$. Converting the sum to an integral over $\ln \ell$ gives $\sum_\ell \frac{2\ell+1}{4\pi} C_\ell \approx \int d\ln\ell\, \frac{\ell(2\ell+1)}{4\pi} C_\ell \approx \int d\ln\ell\, \frac{\ell(\ell+1)}{2\pi} C_\ell$. So

$$D_\ell \equiv \frac{\ell(\ell+1)}{2\pi} C_\ell\, T_0^2 \quad [\mu\mathrm{K}^2] \qquad (7)$$

is the temperature variance per unit $\ln\ell$, with $T_0 = 2.7255 \times 10^6\,\mu$K. A scale-invariant Sachs–Wolfe plateau has $C_\ell \propto 1/[\ell(\ell+1)]$, so its D_ℓ is flat, which is why the plots use D_ℓ. Example: $C_\ell = 10^{-10}$ at $\ell = 10$ gives $D_\ell = 17.5 \times 10^{-10} \times 7.43 \times 10^{12} = 1.30 \times 10^4\,\mu\mathrm{K}^2$. In code, D_ℓ is computed from the map with `healpy.anafast` after removing the monopole and dipole. The dipole is removed because the observer's motion dominates it.

Cosmic variance. If the $a_{\ell m}$ are Gaussian, $(2\ell+1)\hat{C}_\ell/C_\ell$ is χ^2-distributed with $2\ell+1$ degrees of freedom (the m modes are real combinations). That distribution has variance $2(2\ell+1)$, so

$$\frac{\Delta C_\ell}{C_\ell} = \sqrt{\frac{2}{2\ell+1}}. \qquad (8)$$

Values: 63% at $\ell = 2$, 31% at 10, 18% at 30, 6.7% at 220, 3.2% at 1000. This is the irreducible scatter of one sky about its ensemble mean, and it is the $\pm 1\sigma$ band drawn around every CoE curve.

Validity. It is exact for a Gaussian, isotropic, full-sky field. It underestimates the scatter for non-Gaussian skies, e.g. a few dominant voids or the axisymmetric CoE background, and for masked skies. It does not apply to a deterministic component, which has no ensemble scatter at all.

Primers and the inverse problem. An axisymmetric pattern ($m = 0$) is a sum of Legendre polynomials $P_\ell(\cos\theta)$, so a single P_ℓ puts all its power in one ℓ (Fig. 5). N equal masses equally spaced on a ring have azimuthal Fourier power only at multiples of N (aliasing). Binary mixtures and quadrature masses behave the same way. *Inverse problem*: given a target C_ℓ, find over-/under-densities that produce it. Because C_ℓ discards the phases of the $a_{\ell m}$, the solution is degenerate (Figs. 5–6).

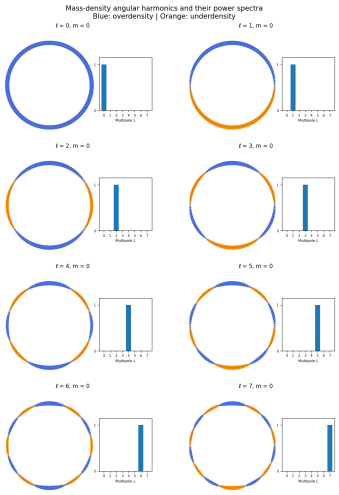


Figure 5: Single axisymmetric harmonics $\ell = 0 \ldots 7$ drawn as density on a ring (blue over-, orange under-density) with the power bar at that ℓ.

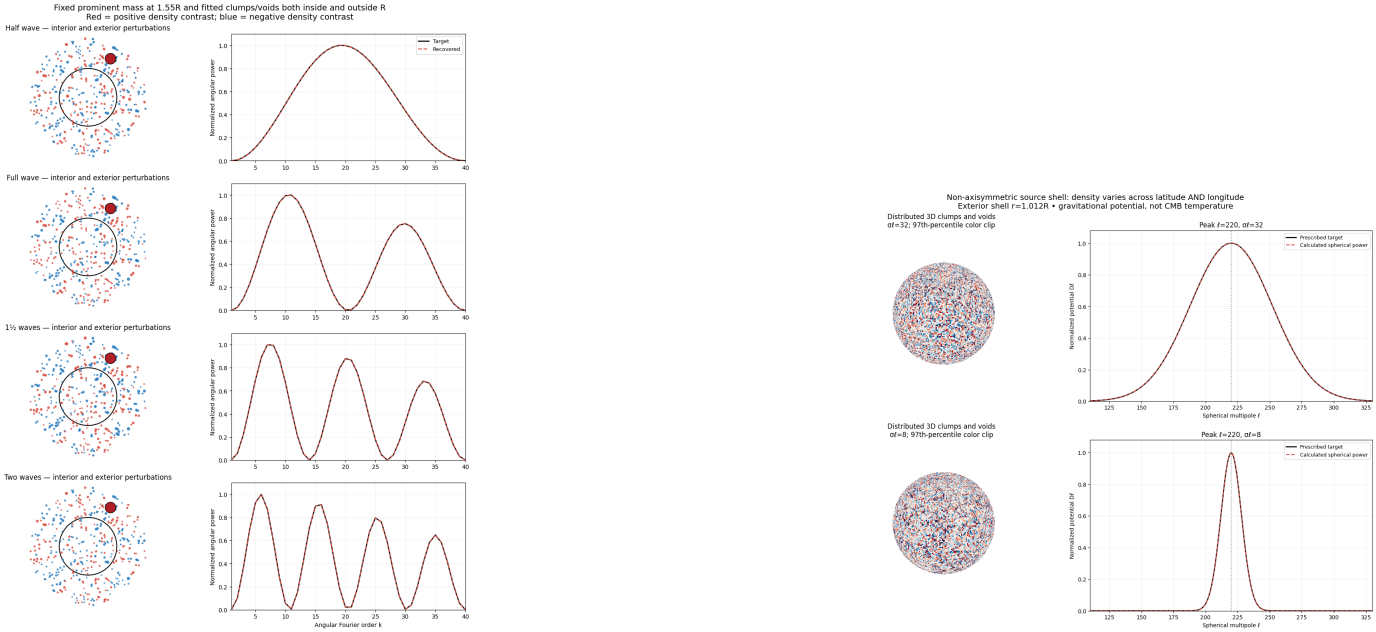


Figure 6: Inverse problem solved numerically. Left: a fixed mass at $1.55R$ plus fitted clumps/voids reproduce target angular profiles (half, $1\frac{1}{2}$, $2\frac{1}{2}$, two waves). Right: a non-axisymmetric source shell at $r = 1.012R$ (gravitational potential, not temperature) built from 3D clumps/voids reproduces a prescribed peak at $\ell = 220$ with widths $\sigma_\ell = 32$ and 8.

4 Goodness-of-fit statistics

χ^2. Inputs: data d_ℓ, model m_ℓ, Planck errors. Planck low-ℓ errors are asymmetric (σ_ℓ^-, σ_ℓ^+), so the code uses the error on the side where the model lies:

$$\chi^2 = \sum_\ell \frac{(m_\ell - d_\ell)^2}{\sigma_\ell^2}, \quad \sigma_\ell = \begin{cases} \sigma_\ell^+ & m_\ell > d_\ell \\ \sigma_\ell^- & m_\ell \le d_\ell \end{cases}. \tag{9}$$

Meaning: each term is the squared distance in units of measurement error. For N independent Gaussian points and a correct model, $\langle \chi^2 \rangle \approx N - p$, with p the number of fitted parameters. Example: ΛCDM over ℓ 2–30 (29 points) gives 20.1. Here χ^2 uses Planck errors only, because Planck's errors already include cosmic variance for ΛCDM. The stochastic-field inverse fits also used $\chi^2_{\rm CV}$, which adds the model's own cosmic variance $\frac{2}{2\ell+1} D^2_{\rm field}$ in quadrature.

Validity. It assumes independent Gaussian errors. It fails at low ℓ, where the likelihood is skewed (hence the asymmetric errors), and for the binned $\ell \ge 30$ file, where bins are not fully independent. Read χ^2 as a comparison measure, not an exact probability.

Pearson r^2. With means $\bar{d}$, $\bar{m}$,

$$r^2 = \frac{\left[\sum (d_\ell - \bar{d})(m_\ell - \bar{m})\right]^2}{\sum (d_\ell - \bar{d})^2 \sum (m_\ell - \bar{m})^2}. \tag{10}$$

It is the squared linear correlation and is invariant under $m \to am + b$. It measures shape agreement and ignores amplitude and offset; it is *not* a fit R^2. Example: ΛCDM at ℓ 2–30 has $r^2 = 0.012$ although $\chi^2 = 20.1$, because the data there are scatter around a nearly flat curve. Over ℓ 2–2500, where the peaks dominate the variance, ΛCDM has $r^2 = 0.983$. So r^2 is informative only where the signal varies a lot compared with the noise.

5 A realistic web of voids and clusters

Purpose. Hand-placed masses are replaced by a statistically realistic population. Each body sources a Newtonian potential, and the sky signal is the gravitational field sampled on spheres around the observer.

Ingredients (from `realweb/build.py`, `cosmo.py`). Background: $h = 0.7$, $\Omega_m = 0.31$, $\Omega_b = 0.049$, $n_s = 0.965$, $\sigma_8 = 0.8$, $\bar\rho_m = \Omega_m\, 2.775 \times 10^{11} h^2 = 4.215 \times 10^{10}\, M_\odot \mathrm{Mpc}^{-3}$. Linear spectrum $P(k) = A k^{n_s} T^2_{\rm EH}(k)$ (Eisenstein–Hu no-wiggle transfer function). A is fixed by $\sigma^2(R) = \int \frac{k^2 dk}{2\pi^2} P(k) W^2(kR) = 0.8^2$ at $R = 8/h$ Mpc, with the top-hat $W(x) = 3(\sin x - x\cos x)/x^3$.

Clusters, Press–Schechter. A region of mass $M = \frac{4}{3}\pi R^3 \bar\rho_m$ collapses when its linear overdensity exceeds $\delta_c = 1.686$. The fraction above threshold is $\mathrm{erfc}(\nu/\sqrt{2})$ with $\nu = \delta_c/\sigma(M)$; differentiating with respect to $\ln M$ and multiplying by $\bar\rho_m/M$ gives

$$\frac{dn}{d\ln M} = \sqrt{\frac{2}{\pi}} \frac{\bar\rho_m}{M} \nu \left| \frac{d\ln\sigma}{d\ln M} \right| e^{-\nu^2/2}, \qquad (11)$$

for $M = 10^{14}$–$2 \times 10^{15} M_\odot$ (41 bins). The peak-background bias is $b = 1 + (\nu^2 - 1)/\delta_c$: halos are more numerous where the large-scale $\delta_{\rm lin} > 0$.

Voids, Sheth–van de Weygaert (2004). This is the two-barrier excursion-set size function with void threshold $\delta_v = -2.7$, evaluated at linear radius $R_L = R/1.7$ for $R = 10$–50 Mpc. Bias -1.5. A void of radius R_v has interior contrast -0.85, i.e. mass deficit $M_d = 0.85 \bar\rho_m \frac{4\pi}{3} R_v^3$. Example: $R_v = 30$ Mpc gives $M_d = 4.0 \times 10^{15} M_\odot$. It is painted as $-M_d$ at the centre plus 8 random points of $+M_d/8$ on a shell at $1.25 R_v$, so each void is compensated (net mass zero).

Placement. Counts per cell are Poisson with mean

$$\lambda = \frac{dn}{d\ln X} \Delta\ln X\, \Delta V\, \max(0, 1 + b\,\delta_{\rm lin})\, m(\mathbf{x})\, f, \qquad (12)$$

with $X = M$ or R_v, thinning $f = 1/100$, and $m(\mathbf{x})$ the CoE modulation of Fig. 8. The painted ρ is mean-subtracted inside the region (zero net mass), then Φ is found by FFT Poisson (§6).

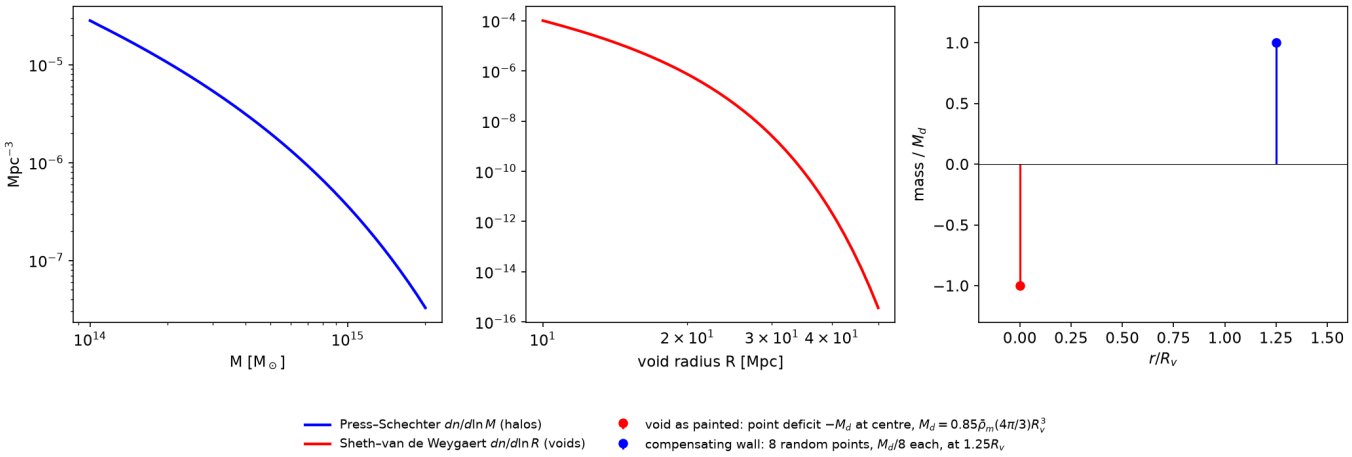


Figure 7: Web ingredients recomputed from `cosmo.py`: halo mass function (left), void size function (middle), compensated void mass placement (right).

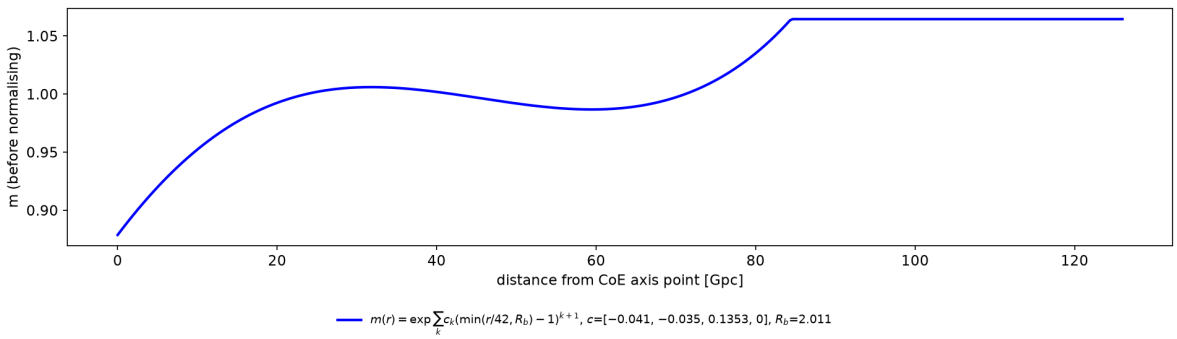


Figure 8: CoE modulation $m(r) = \exp\sum_k c_k(\min(r/42, R_b) - 1)^{k+1}$ of the web number density. r is measured from the axis point $42\hat{a}$ Gpc, $\hat{a} \propto (1, 0, \tan 15°)$; $c = [-0.041, -0.035, 0.1353, 0]$, $R_b = 1.5(1 + 1/2.937)$. Normalised to mean 1 in each grid shell.

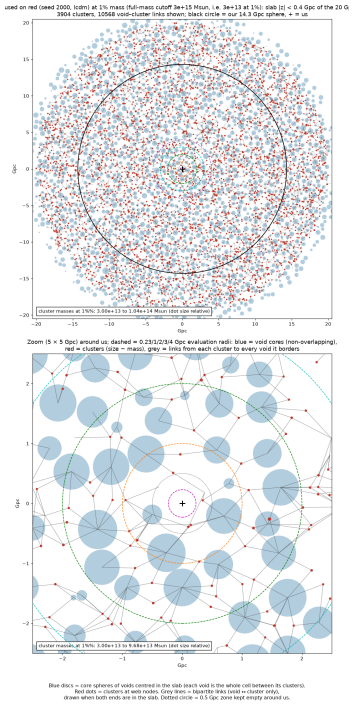


Figure 9: Realistic web: slab $|z| < 0.4$ Gpc of a 20 Gpc box (seed 2000, 1% mass cut, 3904 clusters, 10568 void–cluster links). Circle: 14.3 Gpc sphere; +: observer. Bottom: zoom (±9 Gpc). [unverified: whether this figure used the same `build.py` as above (whose grids use seeds 11, 12)].

The kernel ladder: what "kernel at R" means. From `realweb/build.py` (`skyfrom`) and `mc/mc.py`, `mc_samples.py`. *Step 1, acceleration*: $\mathbf{a} = -\nabla\Phi$ on the grid, from Poisson in Fourier space, $\mathbf{a}_k = -i\mathbf{k}\,(4\pi G\rho_k/k^2)$, with a Gaussian of cell width applied to represent finite body size. *Step 2, radial tidal signal on a sphere*: for each HEALPix direction $\hat{n}$ ($N_{\rm side} = 32$, 12,288 directions) and each radius $R \in \{0.5, 0.9, 2, 4, 8, 14\}$ Gpc,

$$s_R(\hat{n}) = \big[\mathbf{a}(R\hat{n}) - \mathbf{a}(\mathbf{0})\big] \cdot \hat{n} - \langle\cdot\rangle_{\rm sky}. \qquad (13)$$

This is the radial component of the acceleration at distance R *relative to the observer's own acceleration*: a tidal (differential) field, since a uniform acceleration is unobservable. "Kernel at R Gpc" therefore means that the web's sky imprint is evaluated on the sphere of radius R. Small R weights nearby structure; large R samples the field near the source surface. The map is multiplied by $M_\odot/K_G \cdot R_V^2$ ($K_G = 7.7 \times 10^{51}/0.0039$, $R_V = 14.3$) to put it in the same dimensionless units as the deterministic "red" map. [unverified: the physical origin of the constant K_G]. *Step 3, Monte-Carlo universe*: each universe has a new Gaussian $\delta_{\rm lin}$ and Poisson halos in 8 mass bins (voids omitted in the fast variant, $N = 256$ grids, seeds $500000 + i$). *Step 4, the % levels*: the sky analysed is

$$\Theta_\epsilon(\hat{n}) = \Theta_{\rm red}(\hat{n}) + \epsilon\, s_R(\hat{n}), \qquad (14)$$

with $\Theta_{\rm red}$ the smooth axisymmetric CoE background (`sm_red`, $\Theta_{\rm red} = e^{-S(\psi)}/\langle e^{-S}\rangle - 1$, with S a spline in angle ψ from the axis). For each $\epsilon \in \{100, 50, 20, 10, 5, 2, 1, 0.5, 0.2, 0.05\}\%$ the code computes D_ℓ via Eq. (7) and χ^2 via Eq. (9). So "10%" means the web field is scaled to one tenth of its natural (strength-1) amplitude before being added to the background; since D_ℓ is quadratic in the map, its web part scales as ϵ^2. The "best" curve minimises χ^2 over 60 log-spaced ϵ from $10^{-3.5}$ to $10^{0.3}$ (0.03%–200%). *Reading the ladder*: as $\epsilon \to 0$ the curves converge on the smooth-red spectrum (dashed red, $\chi^2 = 471.4$ in the figure legends), whose power sits in odd ℓ because it is axisymmetric. Increasing ϵ fills in the even ℓ and raises the overall level.

Labels on Fig. 10. The panel titles carry "my guess: PASS/FAIL" labels written by `mc_samples.py`. Its rule is: PASS if $\chi^2_{\rm best} \le 100$, |slope of $\ln D_\ell$ vs ℓ over ℓ 4–30| < 0.05, and $\max D_{2..4}/\max D^{\rm Planck}_{2..4} < 2.5$. These are the script's labels, reproduced in the figure; no verdict is drawn here. Recorded values for universe #3 (`samples/guesses.json`), as R: $\epsilon_{\rm best}$, χ^2: 0.5: 29.0%, 441.6; 0.9: 25.0%, 411.5; 2: 25.0%, 177.2; 4: 21.6%, 69.6; 8: 52.5%, 85.0; 14: 45.3%, 47.2. [unverified: at 8 Gpc the JSON records the low-ℓ ratio 2.99, which exceeds the coded 2.5 threshold, yet the label is PASS: the JSON and the script as now written disagree].

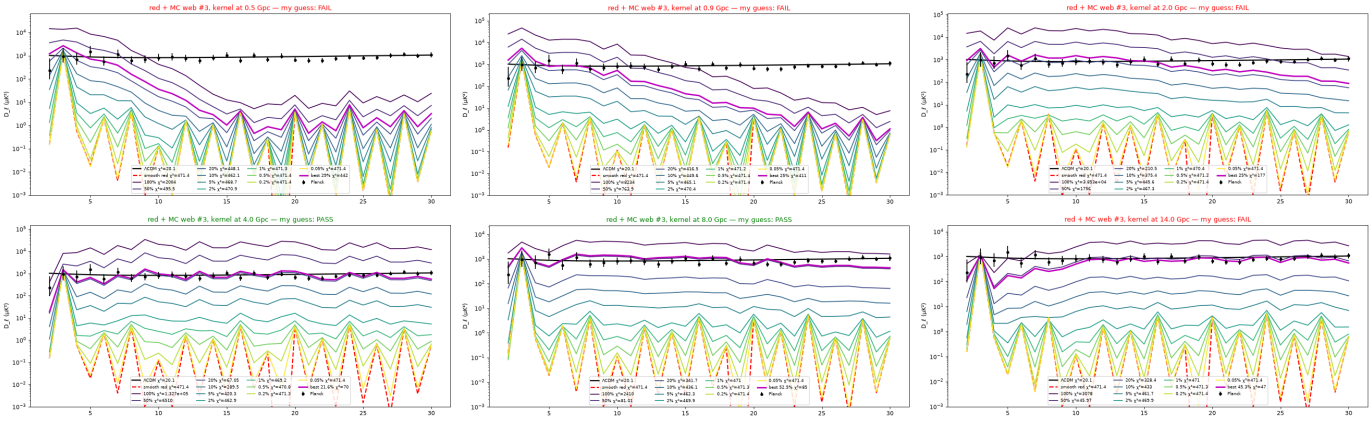


Figure 10: Red + Monte-Carlo web #3, kernel at $R = 0.5$, 0.9, 2.0 (top) and 4, 8, 14 Gpc (bottom). Panels are rearranged from the original single column. Coloured curves: D_ℓ of $\Theta_{\rm red} + \epsilon s_R$ for the listed ϵ; magenta: best ϵ; red dashed: smooth red alone; black line: ΛCDM ($\chi^2 = 20.1$); points: Planck. The title colours and "my guess" labels come from the plotting script.

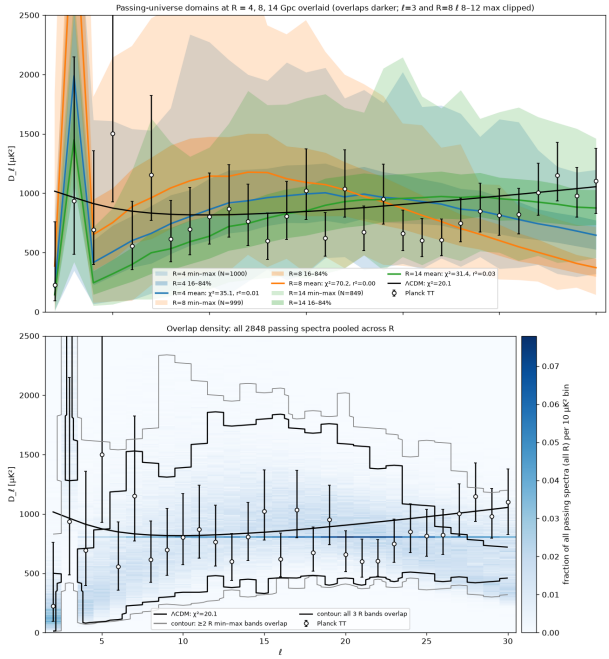


Figure 11: Passing-universe domains at $R = 4$, 8, 14 Gpc overlaid (top; bands are spreads over realisations; $\ell = 3$ and the $R = 8$ ℓ 8–12 maxima clipped) and the density of sky-plane domains (bottom) against Planck.

6 Stochastic field and sky projection

Why a random field. Instead of placing objects, the density contrast $\delta(\mathbf{x}) = \rho/\bar{\rho} - 1$ is treated as a zero-mean Gaussian random field. In Fourier space $\langle \delta_{\mathbf{k}} \delta^*_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}') P_\delta(k)$, where P_δ [Gpc^3] is the variance per mode. The model predicts the *ensemble mean* C_ℓ; one sky scatters about it by Eq. (8).

Poisson equation. Newtonian gravity: $\nabla^2 \Phi = 4\pi G \bar{\rho}\, \delta$. Fourier transforming ($\nabla \to i\mathbf{k}$, so $\nabla^2 \to -k^2$),

$$\Phi_{\mathbf{k}} = -\frac{4\pi G\bar{\rho}}{k^2}\delta_{\mathbf{k}} \;\Rightarrow\; P_\Phi = \frac{(4\pi G\bar{\rho})^2}{k^4} P_\delta. \qquad (15)$$

Code units: $G = 4.30 \times 10^{-12}\,\mathrm{Gpc(km/s)^2} M_\odot^{-1}$ and $\bar{\rho} = 4.215 \times 10^{19} M_\odot \mathrm{Gpc}^{-3}$, so $4\pi G\bar{\rho} = 2.278 \times 10^9\,(\mathrm{km/s})^2 \mathrm{Gpc}^{-2}$. Example: a mode with $k = 1\,\mathrm{Gpc}^{-1}$ and $\delta_k = 10^{-4}$ gives $|\Phi| = 2.28 \times 10^5\,(\mathrm{km/s})^2$, i.e. $\Phi/c^2 = 2.5 \times 10^{-6}$. Meaning: the k^{-2} makes the potential dominated by large scales.

Validity. Valid for $|\Phi|/c^2 \ll 1$, non-relativistic sources, and scales well inside the horizon ($kR \gg 1$). At $\ell \lesssim 3$, where $k \sim 1/R$, relativistic (GR) corrections are not neg-

ligible. The code also evaluates the moving lens with the full CoE radial velocity, which is superluminal ($\dot{R} \approx 4c$ today per the stage-A notes); the Newtonian treatment of that regime is a model assumption, not a derived result.

Source terms and the moving lens. A photon crossing a potential picks up (i) Sachs–Wolfe $\Theta = \Phi/3c^2$ at the source, (ii) integrated Sachs–Wolfe $\frac{2}{c^2}\int \dot{\Phi}\, dt$ if Φ evolves, and (iii) the moving-lens term. If the lens moves with transverse velocity $\mathbf{u}$, the potential the photon sees changes during the crossing, giving $\Theta \propto \frac{2}{c^2}\int d\chi\,(\text{time per } d\chi)\,\mathbf{u}\cdot\nabla\Phi$. In `fwd.py` this is the kernel $K_M(\chi) = \frac{2}{c^2}\big[-\frac{\bar{A}}{a}\frac{\Delta t}{\Delta\chi}\,42\,R_d\big]$ multiplying $\mathbf{u}\cdot\nabla\Phi_0$ [(km/s)^2/Gpc]. Here $\bar{A}$ is the growth factor D/a, $\Delta t/\Delta\chi$ the time spent per unit path, $42\,R_d$ the CoE radial-velocity factor tabulated from `flowR.py`, and χ the distance from the observer. The stage-A README reports that the moving lens is $\geq 99.99\%$ of the field's power (SW+ISW $\leq 9\times10^{-5}$), so only K_M is kept in the Limber part. [unverified: the closed form of R_d; it is taken from tabulated `tab.npz`].

The $k^2/3$ factor. $\mathbf{u}\cdot\nabla\Phi \to i(\mathbf{k}\cdot\mathbf{u})\Phi_{\mathbf{k}}$. Averaging over random orientations of $\hat{k}$ relative to $\mathbf{u}$: $\langle(\hat{k}\cdot\hat{u})^2\rangle = \frac{1}{4\pi}\int\cos^2\theta\, d\Omega = \frac{1}{3}$, so the power is multiplied by $u^2k^2/3$, with u absorbed into K_M.

Radial window. The field is confined to a shell, $\Phi(\chi\hat{n}) = W(\chi)\phi_g$, with

$$W(\chi) = \exp\Big[-\frac{(\chi - R_c)^2}{2w^2}\Big], \tag{16}$$

centre R_c and width w [Gpc]. *Why Gaussian*: two parameters, smooth (no edge artefacts in ℓ), and its derivative $W' = -W(\chi - R_c)/w^2$ is analytic. That derivative is needed because $\mathbf{u}\cdot\nabla(W\phi) = W\,\mathbf{u}\cdot\nabla\phi + \phi\,W'(\hat{n}\cdot\mathbf{u})$. Fitted: $R_c = 3.5715$, $w = 0.3355\,$Gpc with $R = 4$. The source surface at 4 Gpc is $(4 - 3.5715)/0.3355 = 1.28w$ above the centre, so the path truncates the Gaussian. That truncation is why Limber needs correcting (below).
Validity. A Gaussian is a parametrisation, not a derived shell profile. It fails to represent sharp-edged or multi-shell structures.

Exact projection (derivation). *Step 1*: the line-of-sight integral is $\Theta(\hat{n}) = \int d\chi\, K(\chi)W(\chi)\,\Phi(\chi\hat{n})$. *Step 2*: Fourier expand, $\Phi(\chi\hat{n}) = \int\frac{d^3k}{(2\pi)^3}\Phi_{\mathbf{k}}e^{i\mathbf{k}\cdot\hat{n}\chi}$, and use the plane-wave expansion $e^{i\mathbf{k}\cdot\hat{n}\chi} = 4\pi\sum_{\ell m} i^\ell j_\ell(k\chi)Y_{\ell m}(\hat{n})Y^*_{\ell m}(\hat{k})$. *Step 3*: read off $a_{\ell m} = \frac{i^\ell}{2\pi^2}\int d^3k\,\Phi_{\mathbf{k}}Y^*_{\ell m}(\hat{k})\Delta_\ell(k)$ with $\Delta_\ell(k) = \int d\chi\, KW j_\ell(k\chi)$. *Step 4*: take $\langle|a_{\ell m}|^2\rangle$ with the power-spectrum definition. The δ^3 collapses one k integral and $\int d\Omega_k|Y_{\ell m}|^2 = 1$, giving

$$C_\ell = \frac{2}{\pi}\int k^2dk\, P_\Phi(k)\,|\Delta_\ell(k)|^2. \tag{17}$$

j_ℓ is the spherical Bessel function: the radial profile of a plane wave's ℓ-th multipole. For the moving lens, $P_\Phi \to \frac{k^2}{3}P_\Phi$, and `fwd.py` evaluates the W' term exactly with $\cos\theta$ Gaunt couplings to $\ell\pm1$. This is used for $\ell \leq 50$.

Limber approximation (derivation). For large ℓ, $j_\ell(k\chi)$ is sharply peaked at $k\chi \approx \nu \equiv \ell + \frac{1}{2}$. Using the closure relation $\int k^2dk\, j_\ell(k\chi)j_\ell(k\chi') = \frac{\pi}{2\chi^2}\delta(\chi - \chi')$, and assuming P_Φ varies slowly over the width of j_ℓ so it can be evaluated at $k = \nu/\chi$, Eq. (17) collapses to a single integral:

$$\begin{aligned} D_\ell \simeq &\frac{\ell(\ell+1)T_0^2}{2\pi}\int d\chi \\ &\times\frac{[K_MW]^2}{\chi^2}\frac{k^2}{3}P_\Phi(k)\Big|_{k=\nu/\chi}. \end{aligned} \tag{18}$$

Meaning: each multipole receives power only from the wavenumber that projects to angle $1/\ell$ at distance χ. Only transverse modes contribute; radial modes cancel along the line of sight.
Validity. It requires the kernel K_MW to vary slowly compared with j_ℓ's oscillation $\sim \chi/\nu$, i.e. $\ell \gg R_c/w = 3.5715/0.3355 = 10.6$, and no sharp edges. Checks in `acoustic_limber_check*.json`: interior or broad shells agree within 3% for $\ell \geq 150$, and a 0.5 Gpc shell within 0.3% even at $\ell = 20$. Shells cut off at the path end (as here) are overestimated by about 1.5–5× below $\ell = 150$ and by up to ±20% near $\ell = 300$.

Exact/Limber correction. Hence $D_\ell^{\rm used} = c_\ell D_\ell^{\rm Limber}$ for $\ell > 50$, with $c_\ell = C_\ell^{\rm exact}/C_\ell^{\rm Limber}$ computed by Eq. (17) at 69 nodes. The nodes are $\ell = 50, 55, 60, 70, 80, 90, 100, 115, 130, 150, 175$, then 200 to 2480 in steps of 40. c_ℓ is interpolated in $\ln\ell$ and recomputed after each fit pass, because it depends on R_c and w.

Fitted source spectrum.

$$\begin{aligned} P_\delta = &10^{\lg A}k^n e^{-(k/k_D)^2}\big[1 + A_w\cos(kr_s + \phi) \\ &+ Be^{-k/k_B}\cos(kr_2 + \phi_2)\big], \end{aligned} \tag{19}$$

with k in Gpc^{-1} and P_δ in Gpc^3. Term by term: k^n is the broadband tilt; $e^{-(k/k_D)^2}$ is a small-scale damping cutoff; $A_w\cos(kr_s + \phi)$ is an oscillation in k with period $2\pi/r_s$, i.e. a preferred comoving separation r_s (acoustic-like); the B term is a second, decaying oscillation. Fitted (r_2 free, `acoustic_half_fixed.json`): $\lg A = -8.282$, $n = -0.183$, $A_w = 0.311$, $r_s = 79.5\,$Mpc, $\phi = 1.367$, $k_D = 335\,$Gpc^{-1}, $B = 0.815$, $k_B = 138\,$Gpc^{-1}, $r_2 = 42.1\,$Mpc, $\phi_2 = 3.640$. Example: at $k = 100\,$Gpc^{-1}, $10^{-8.282}\times100^{-0.183}\times e^{-0.089} = 2.2\times10^{-9}\,$Gpc^3 before the bracket.

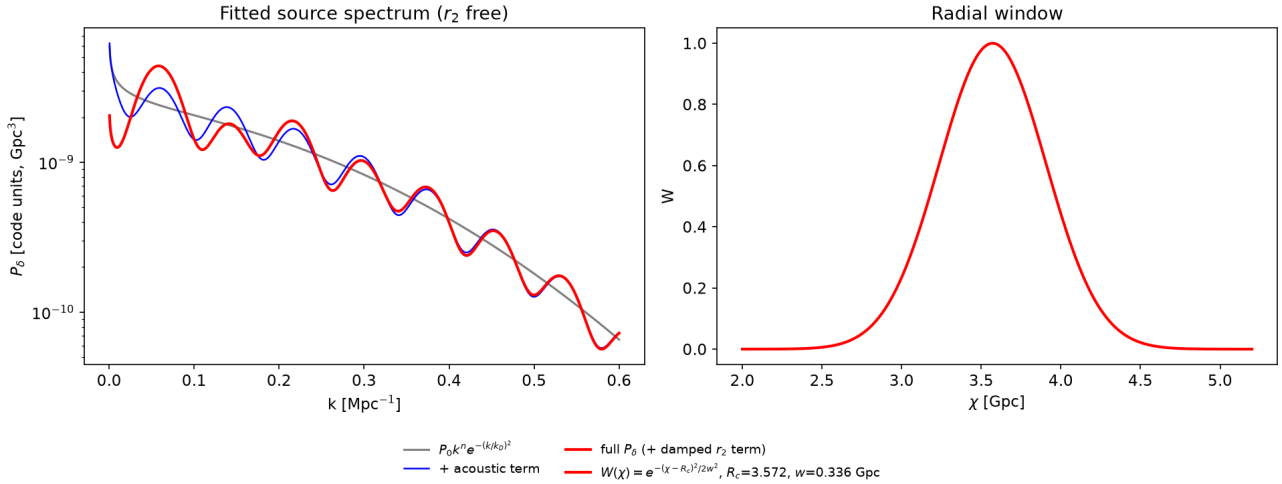


Figure 12: Fitted P_δ split into its terms (left) and the radial window $W(\chi)$ (right).

Mapping $\Delta\ell \leftrightarrow r$ via $\ell \approx k\chi$. From Limber, a wavenumber k at distance χ appears at $\ell \approx k\chi$. An oscillation $\cos(kr_s)$ has period $\Delta k = 2\pi/r_s$, so on the sky its peaks are spaced by

$$\Delta\ell \approx \chi_{\rm eff}\Delta k = 2\pi\chi_{\rm eff}/r_s. \qquad (20)$$

Example: acoustic fit, $R_c = 3.944\,$Gpc, $r_s = 0.0819\,$Gpc, gives $\Delta\ell = 302.6$; the model peaks 227, 530, 831, 1131, 1431 are spaced 300–303. For the final fit, $R_c = 3.5715$ and $r_s = 0.0795$ give 282, whereas the model peaks are spaced ≈ 300. That implies $\chi_{\rm eff} = 300 \times 0.0795/2\pi \approx 3.8\,$Gpc, larger than R_c because K_M rises toward the source and the truncated window is skewed. Conversely, an angular feature at ℓ has physical size $\chi\theta$ with $\theta = \pi/\ell$. For $\ell = 220$: 57.1, 114.2, 199.9 Mpc at $\chi = 4$, 8, 14 Gpc.

Validity. It holds where Limber holds and for thin kernels. It fails at low ℓ and for broad kernels, where many χ contribute and $\chi_{\rm eff}$ is ambiguous.

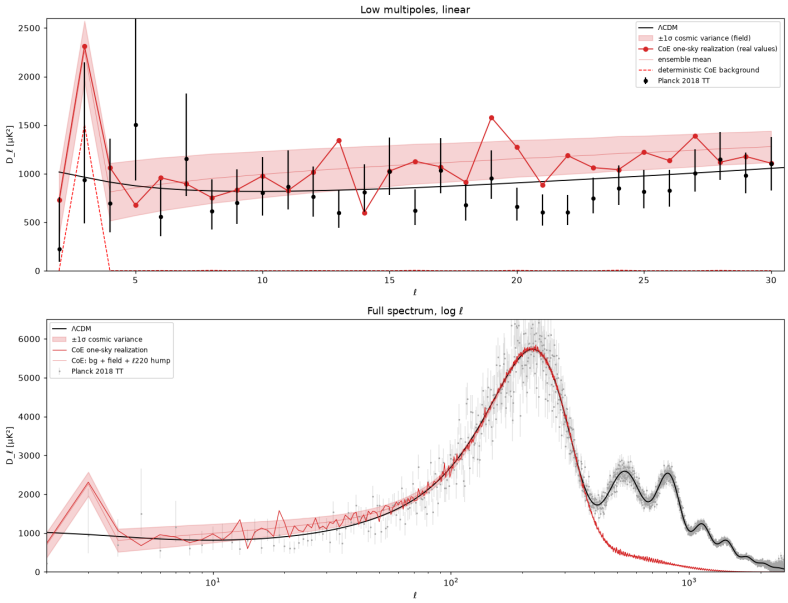


Figure 13: Low multipoles (linear axes): ΛCDM (black), Planck (grey), CoE one-sky realisation (red) in its $\pm1\sigma$ cosmic-variance band (Eq. 8) with faint ensemble mean. Bottom: full spectrum.

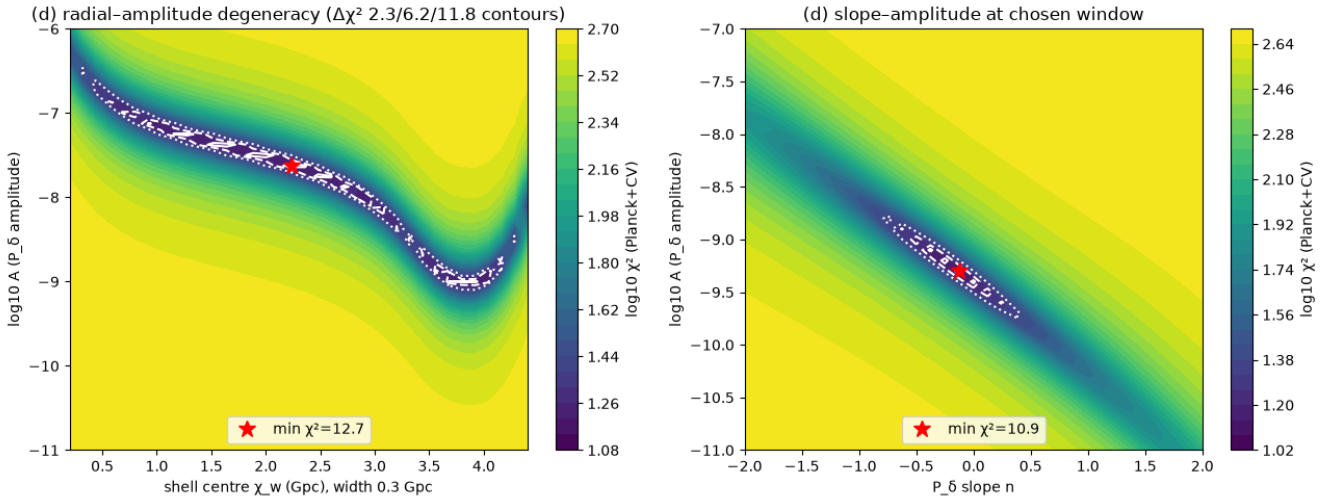


Figure 14: Parameter degeneracies. Left: shell position vs amplitude ($\Delta\chi^2 = 2.3, 6.2, 11.8$, i.e. 1, 2, 3σ for two parameters). A farther shell needs a different amplitude because K_M varies with χ. Right: tilt vs amplitude at the chosen window.

7 The fixed low-ℓ background $D_\ell^{\rm fix}$

Definition (from `fit.py`). The CoE sky has a deterministic, axisymmetric part: the smooth red map plus an `rs_cc` map. Its spectrum is computed once,

$$D_\ell^{\rm fix} = \frac{\ell(\ell+1)}{2\pi} T_0^2\, \hat{C}_\ell\big[\Theta_{\rm red} + \Theta_{\rm rs_cc}\big], \quad 2 \le \ell \le 30, \quad (21)$$

with the dipole removed, and stored in `fit_l30_bg.json`. It is added to the field: $D_\ell^{\rm model} = c_\ell D_\ell^{\rm field} + D_\ell^{\rm fix}$. *Why add powers*: the field has zero mean and is independent of the background, so $\langle|a^{\rm bg} + a^{\rm f}|^2\rangle = |a^{\rm bg}|^2 + \langle|a^{\rm f}|^2\rangle$ (the cross term averages to zero). *Values*: $\ell = 3$: $1506.4\,\mu{\rm K}^2$; $\ell = 2$: 0.1; all others ≤ 5.1, with odd ℓ (5, 7, …) carrying 0.3–5.1 and even $\ell \ge 4$ essentially 0. The odd/even pattern is the axisymmetric signature: an odd-parity profile about the axis projects onto odd Legendre polynomials. It is held fixed (no free parameters) and has no cosmic-variance band, since it is deterministic. Above $\ell = 30$ it is set to zero; its power is already negligible above $\ell = 3$. [unverified: the physical definition of `rs_cc`].

8 Acoustic fit and residual decomposition

Acoustic fit. Eq. (19) without the B term, eight free parameters with R_c and w free ("free_W"). Result: $r_s = 81.9\,{\rm Mpc}$, $R_c = 3.944\,{\rm Gpc}$; $\chi^2_{2-2500} = 11{,}772$, Pearson $r^2 = 0.871$ (ΛCDM: 2,558, 0.983). Peaks: CoE 227, 530, 831, 1131, 1431 vs Planck 224, 535, 821, 1122, 1412.

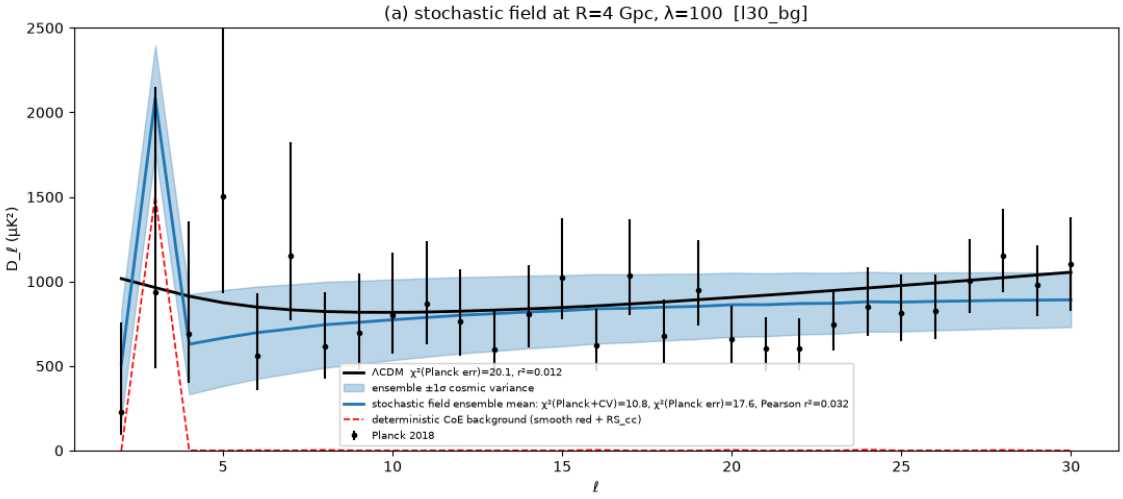


Figure 15: Acoustic stochastic field, $R = 4\,{\rm Gpc}$, free_W: model peaks [227, 530, 831, 1131, 1431] vs Planck [224, 535, 821, 1122, 1412]; red band $\pm1\sigma$ cosmic variance; black ΛCDM.

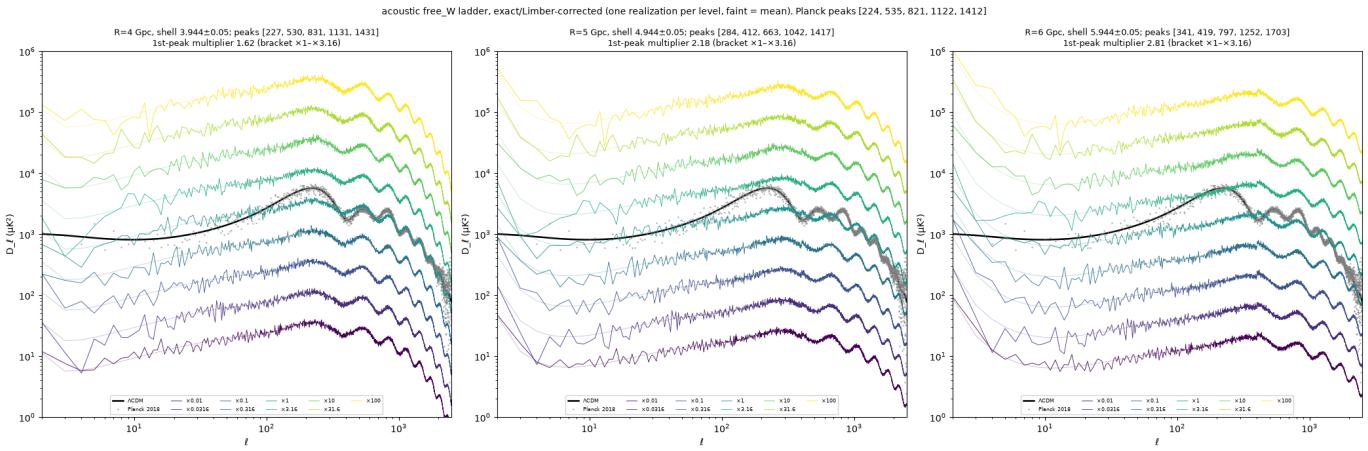


Figure 16: The same source parameters with $R = 4$, 5, 6 Gpc (shell at 3.944, 4.944, 5.944 Gpc): peaks [227, 530, 831, 1131, 1431], [284, 412, 663, 1042, 1417], [341, 419, 797, 1252, 1703]. Moving the shell without refitting r_s changes $\chi_{\rm eff}$ in Eq. (20).

Residuals. Normalised residual: $\epsilon_\ell = (D_\ell^{\rm P} - D_\ell^{\rm CoE})/\sigma_\ell$, dimensionless (in σ). Two linear models with $s_j(\ell) = \sin(2\pi j\ell/\Delta\ell + \phi_j)$:

$$\text{undamped: } \epsilon_\ell = \textstyle\sum_j a_j s_j(\ell), \tag{22}$$

$$\text{damped: } \epsilon_\ell = e^{-\ell/L} \textstyle\sum_j a_j s_j(\ell). \tag{23}$$

Why harmonics: if the model's peak spacing or shape is slightly off, the residual is periodic in ℓ with the spacing error. $j = 2$ is the half period, which is the signature of an odd/even peak alternation. *Why damping*: σ_ℓ shrinks and the signal falls with ℓ, so the misfit in σ units decays. A single envelope $e^{-\ell/L}$ adds one parameter. *Fit*: for fixed $(\Delta\ell, L)$ the model is linear in $a_j \cos\phi_j$, $a_j \sin\phi_j$ (least squares), and $\Delta\ell$, L are scanned. Results: undamped plateaus at $r^2 = 0.551$; damped, 1–9 harmonics give $r^2 = 0.797$, 0.813, 0.831, 0.834, 0.835, 0.836, 0.836, 0.837, 0.837, with $\Delta\ell \approx 593$ and $L \approx 990$. Starting amplitudes (σ): 34.6, 5.6, 5.7, 2.2, 1.3, 1.0, 1.2, 0.4, 0.5. [unverified: these values were taken from the earlier figure set; the script was not re-run]. The decomposition is purely descriptive. It fails where the residual structure is not stationary, e.g. the late troughs at $\ell \sim 1400$, 2000, which the shared envelope has already suppressed.

Mapping to source parameters. Using Eq. (20) with $\chi \approx 3.94$ Gpc: a residual period $\Delta\ell = 593$ corresponds to $r_2 = 2\pi \times 3.94/593 = 0.0417$ Gpc $= 41.7$ Mpc $\approx r_s/2$. An envelope $e^{-\ell/L}$ with $\ell = k\chi$ becomes $e^{-k\chi/L}$, so $k_B = L/\chi = 990/3.94 = 251\,\text{Gpc}^{-1} = 0.25\,\text{Mpc}^{-1}$. These were the starting values for the damped term.

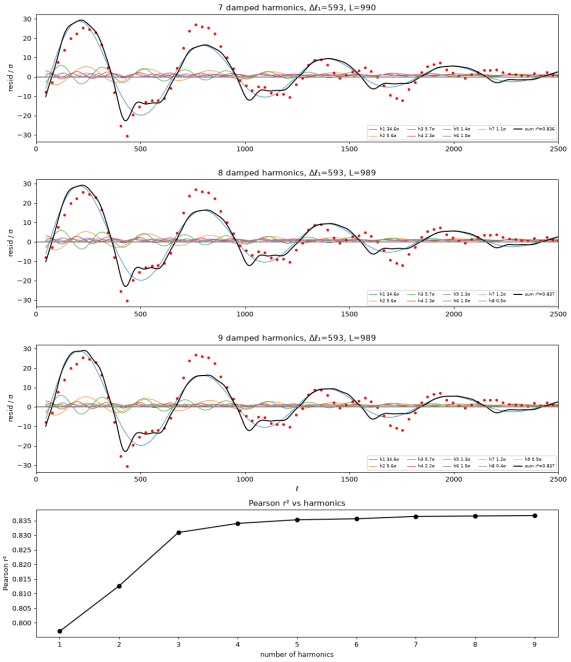


Figure 17: Damped-harmonic fits to ϵ_ℓ with 7, 8, 9 harmonics ($\Delta\ell = 593$, $L = 990$) and Pearson r^2 versus number of harmonics (bottom).

9 Damped half-scale model

Model and fit. Eq. (19) with all 12 parameters $(\lg A, n, A_w, r_s, \phi, k_D, R_c, w, B, k_B, r_2, \phi_2)$ floated in one χ^2 minimisation over $\ell = 2$–2500 (Eq. 9). The "half" variant ties $r_2 = r_s/2$ (11 parameters). Result (r_2 free): $r_2 = 42.1\,\mathrm{Mpc}$ vs $r_s/2 = 39.8$; $k_B = 0.138\,\mathrm{Mpc}^{-1}$, against the starting 0.25.

The $\ell = 50$ step. A ratio c_ℓ computed for an earlier, narrower shell had been left in place when the fit widened the shell, which produced a step at $\ell = 50$ (reported as 43%). Recomputing c_ℓ on the 69 nodes at each pass reduces the 50→51 change to ≈ 1.1%. [unverified: the 43% and 1.1% figures come from the run log; not re-measured here].

Table 1: Fit statistics (Planck errors). Pearson r^2 over ℓ 2–2500.

Model	χ^2_{2-2500}	r^2	χ^2_{2-30}
CoE r_2 free	3,787	0.976	137
CoE half ($r_2 = r_s/2$)	4,131	0.966	49.1
CoE acoustic (no B)	11,772	0.871	–
ΛCDM	2,558	0.983	20.1

peak	Planck	CoE r_2 free	CoE half
1	228 (5726)	218 (5927)	207 (5223)
2	544 (2580)	531 (2449)	537 (2383)
3	823 (2525)	813 (2466)	818 (2433)
4	1121 (1239)	1130 (1293)	1132 (1260)
5	1417 (814)	1420 (822)	1426 (861)

Right: peak ℓ (D_ℓ in $\mu\mathrm{K}^2$).

Remaining structure. The odd/even alternation is absent from the new residual. What remains: a slow wave with $\Delta\ell \approx 1{,}384$ (± 8–11σ), and $\ell = 2$–3 raised to $\approx 3{,}300\,\mu\mathrm{K}^2$, which dominates $\chi^2_{2-30} = 137$; part of the $\ell = 3$ level is $D_3^{\mathrm{fix}} = 1506$. [unverified: the $\Delta\ell \approx 1384$ amplitude and the $3300\,\mu\mathrm{K}^2$ level were read from the earlier figures].

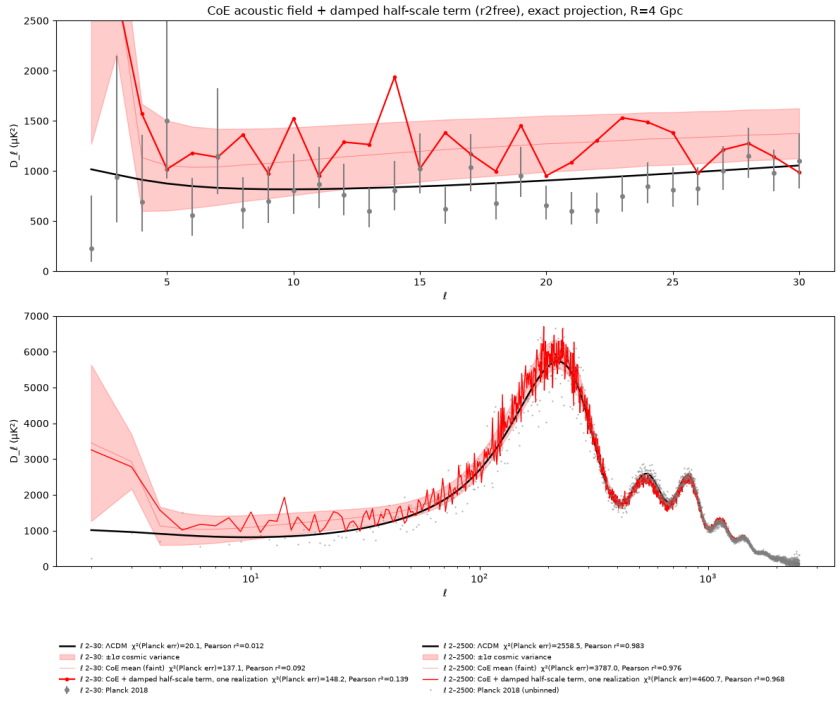


Figure 18: Final model: acoustic field + damped half-scale term (r_2 free), exact/Limber-corrected projection, $R \approx 4\,\mathrm{Gpc}$, plus D_ℓ^{fix}. Top: low ℓ (linear). Bottom: full spectrum. ΛCDM black, Planck grey, CoE one-sky realisation red in its $\pm 1\sigma$ band. $\chi^2_{2-2500} = 3{,}787$, Pearson $r^2 = 0.976$.

10 Parameters and status

D, d	observer–CoE distance 58.2; vision radius 14.3005 Gpc
R, χ	source-sphere radius; distance from observer
a_{eq}	equality scale factor, 1/3400
L, g	density scale length, shape
ϵ, s_R	web amplitude fraction; tidal sky at R
$\lg A$, n	source amplitude, tilt
A_w, r_s, ϕ	acoustic amplitude, scale, phase
k_D	damping cutoff
R_c, w	window centre, width

Predicted vs fitted. Fitted: D, the density-law shape and l_0 (Migkas); the 12 source and shell parameters (CMB). Derived given those: the two-hump Migkas shape, the CatWISE-locked L, and the full D_ℓ via Eqs. (17)–(18). The data require a second damped mode at $r_2 \approx 42\,\mathrm{Mpc} \approx r_s/2$ decaying at $k_B \approx 0.14\,\mathrm{Mpc}^{-1}$; a physical shell model would have to supply it.

Next steps. (1) Exact projection throughout, and control of low ℓ in the r_2-free fit. (2) An analytic thin-shell projection to see which parameters the geometry fixes. (3) $D(t)$ from the Newtonian integrator. (4) Reconcile the PASS label at 8 Gpc with the coded rule.